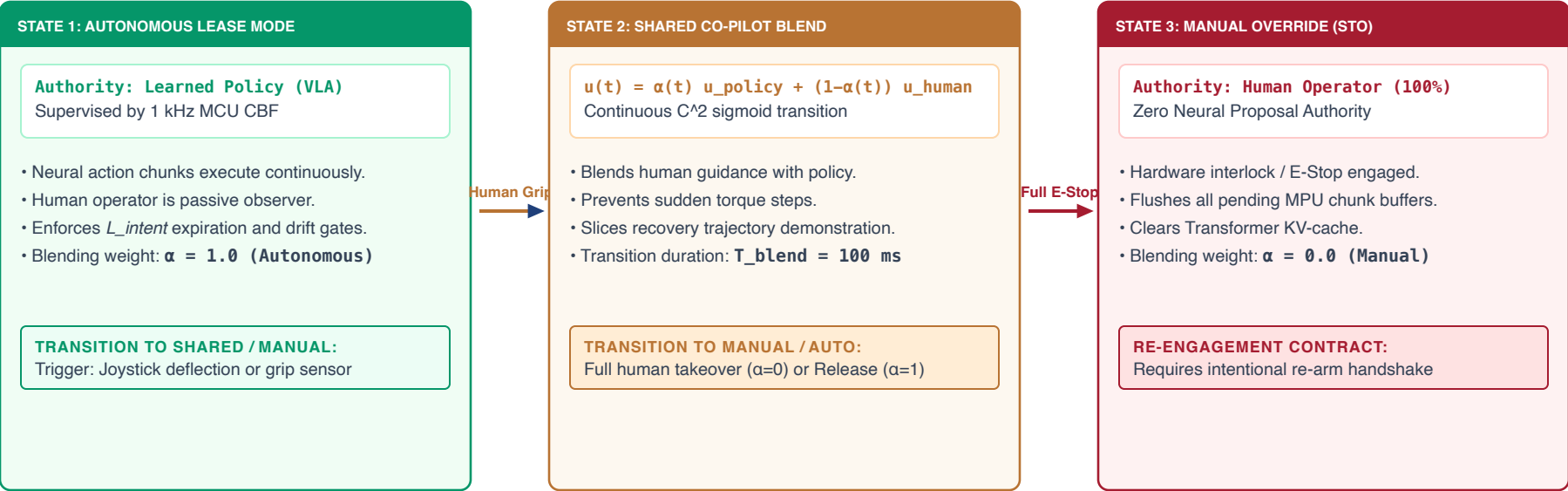


DETERMINISTIC HUMAN–MACHINE AUTHORITY STATE MACHINE

Stateful arbitration between Autonomous Intent Lease, Shared Co-Pilot, and Manual Operator Override



THE PHYSICAL AI GOVERNANCE INVARIANT

- Zero-Software Hardware Preemption:** Physical E-stop buttons and safety lid switches cut gate power directly at the hardware layer.
- Bumpless Transition Guarantee:** Motor torque transitions follow continuous sigmoid blends ($\dot{u} \leq 11\text{ Nm/s}$), preventing gear stripping.
- Cryptographic Provenance:** Every takeover event is signed with monotonic hardware timestamp and SHA-256 telemetry hash.